

Scale-Space Blob Detectors

Audio processing, Video processing and Computer Vision

Lab exercise 2

Introduction

The goal of this lab exercise is to implement a scale-space blob detector using the Laplacian of Gaussian (LoG) filter.

Exercise 1: Non-maxima suppression

Before diving into the development of the scale-space blob detector, let's first explore the non-maximum suppression algorithm and how to implement it.

Follow this pseudocode:

1.- Define a 2-D coordinate grid with dimensions 100×100 , where both the x and y coordinates range from -10 to 10 . Use `np.meshgrid`.

2.- Design a function to create a 2D Gaussian blob:

```
def gaussian(x, y, x0, y0, sigma):
```

where $(x0, y0)$ represents the center of the Gaussian and σ the standard deviation.

3.- Generate an image by combining three Gaussian blobs with standard deviations of 1.2, 1.6, and 2.2. The Gaussians should be centered at:

$(x0=-4, y0=-4)$ with a height of 5,
 $(x0=4, y0=-4)$ with a height of 10, and
 $(x0=0, y0=4)$ with a height of 8.

4.- Non-Maximum Suppression using `rank_filter`:

- Apply `rank_filter` to the image with a filter size of 3 and `rank=-1` and visualize the result.

```
max_filter_output = rank_filter(image, rank=-1, size=3)
```

- Combine the original image with the output from `rank_filter` to retain only the local maxima and suppress all other values.

Display the original image, the `rank_filter` output, and the result after non-maximum suppression.

Exercise 2

To begin, we will apply a Laplacian of Gaussian (LoG) filter to the image "Sunflowers.jpg". Adjust the filter's variance to match the size of the largest sunflowers in the image. The pseudocode for this process is as follows:

```
Filter the image using the LoG filter
Apply thresholding to the filtered image
Perform non-maximum suppression
Display the results
```

Here are a few hints to assist you:

- Use `scipy.ndimage.gaussian_laplace` to implement LoG filter.
- Visually inspect selected detections by reviewing the image at each processing stage: filtering, thresholding, and non-maximum suppression.
- To display the results, you may use the `plot_circles (image, cx, cy, rad)` function, which is included in the materials.

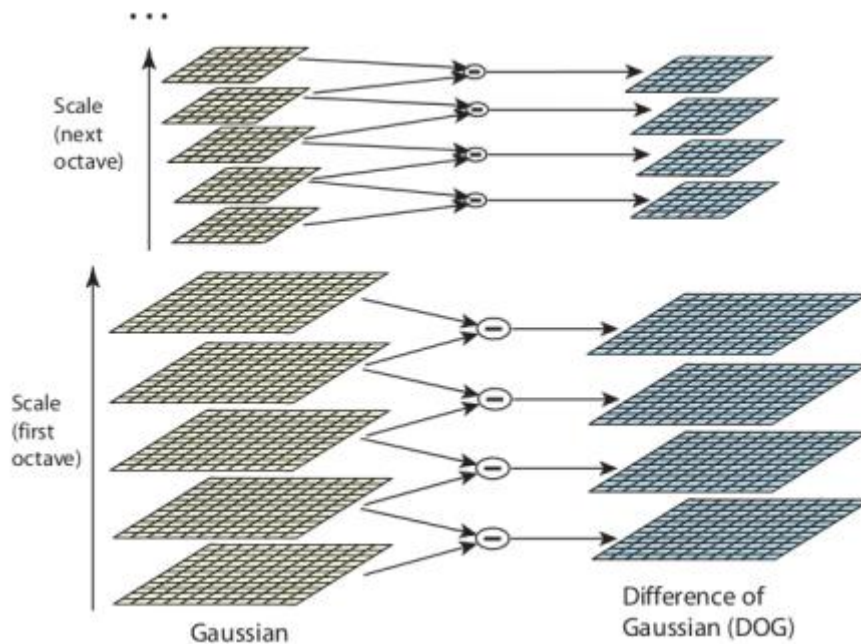
Exercise 3

Now we begin constructing the components of the scale-space blob detector. Following the SIFT approach (see Figure), we define scales in two stages. Within each octave, the scale is progressively increased until it doubles, after which the image is downsampled.

Specifically, the scales are given by

$$\sigma(o, s) = \sigma_0 2^o k^s, \text{ with } s = 0, 1, \dots, S - 1$$

Where o denotes the octave index and $k = 2^{1/s}$.



You can follow this high-level script structure:

```
# Build the scale-space
Define  $\sigma_0$ ,  $N_o$  (number of octaves), and  $S$ 

for o in octaves

    Resize the image

    for sc in scales
        Filter image with the corresponding LoG filter
        Threshold and 2D non-maxima suppression
        3D non-maxima suppression within each octave

# Displaying the results
Display the results
```

Keep using the image “Sunflowers.jpg”.

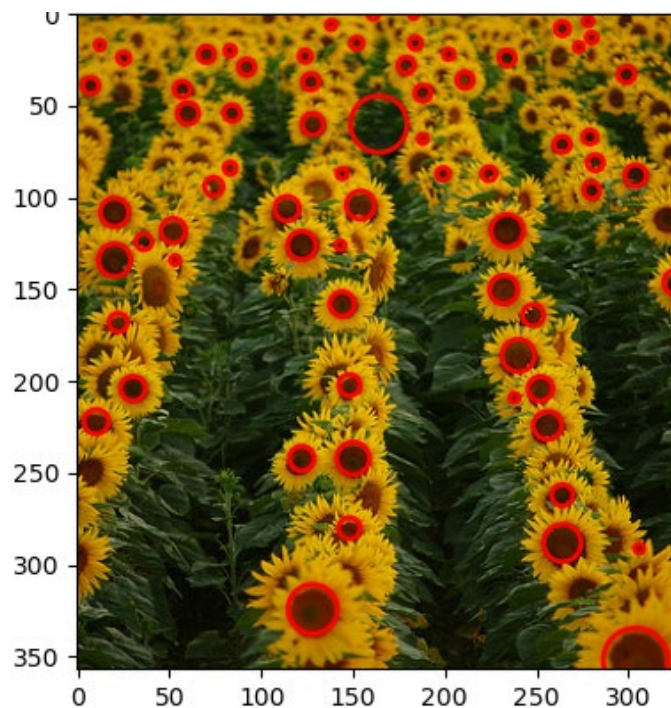
Here are a few hints to assist you:

Build the scale-space

- Explore the size of the sunflowers in the image and propose an appropriate scale-space. Choose σ_0 , N_o and S such that you can detect the smallest and the largest sunflower with a reasonable number of scales (10-15).
- For resizing the images, you can use the function `skimage.transform.resize`.
- Remember to normalize the filter to ensure the filter responses are comparable across different scales.
- To perform the 3D non-maxima suppression within each octave, you can rely again on the function `scipy.ndimage.rank_filter`.

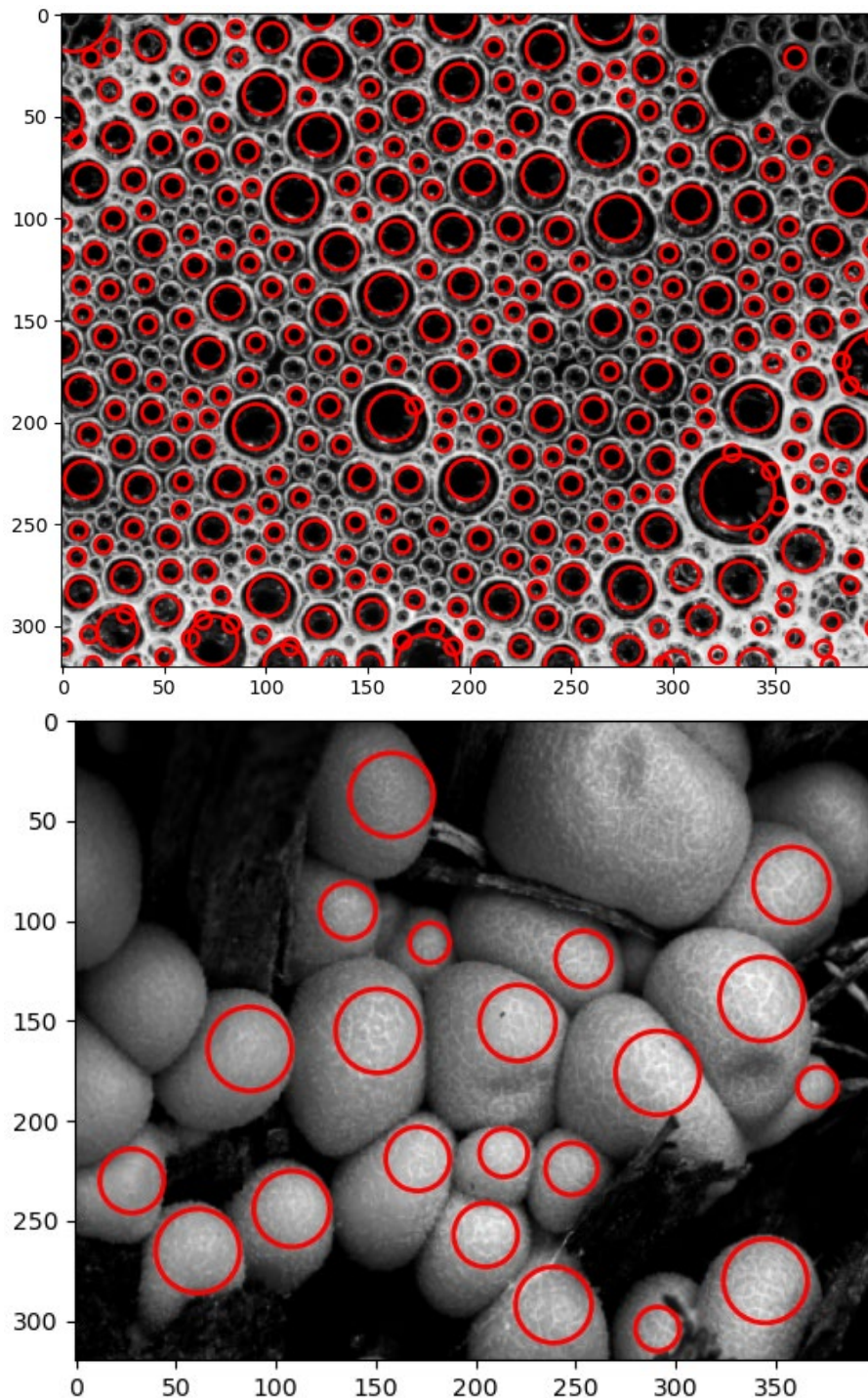
Displaying the results

Here you have an example of how the results may look like. Try to select the parameters of the algorithm to achieve a similar result.



Exercise 4

Now, let's work with two additional images: `fruits.jpg` and `water_texture.jpg` (both photos from [PxHere](#)). How would you choose the parameters of the algorithm to get these results? The parameterization should be different for each image.



Evaluation

The evaluation for this lab will take place on **October 20th (G96)** and **October 22nd (G97)**. You will be required to complete a 15-minute quiz related to this lab.